

Karran Pandey

Curriculum Vitae

Dynamic Graphics Project
University of Toronto
karranpandey.github.io
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RESEARCH INTERESTS

My research builds novel creative controls for 3D-aware visual content creation, editing and exploration. Spanning diverse artistic mediums, my projects range from accelerating traditional workflows in sketch-based design and 3D geometric modeling to building new approaches for controllable photorealistic creation with 3D Gaussian splats and generative image / video models. I aim to make it convenient, intuitive and fun to manifest our visual imagination.

EDUCATION

University of Toronto	2021 - 2025
PhD Computer Science	
Advisor: Karan Singh	Toronto, Canada
Birla Institute of Technology and Science Pilani	2014 - 2019
MSc Mathematics, BE Computer Science	
Advisors: Vijay Natarajan, Tathagata Ray, Sharan Gopal	Hyderabad, India

PUBLICATIONS

Painting with 3D Gaussian Splat Brushes	2025
Karran Pandey , Anita Hu, Or Perel, Clement Fuji-Tsang, Karan Singh, Masha Shugrina ACM SIGGRAPH North America.	
Motions Modes: What Could Happen Next?	2025
Karran Pandey , Matheus Gadelha, Yannick Hold-Geoffroy, Karan Singh, Niloy Mitra, Paul Guerrero CVPR.	
Diffusion Handles: Enabling 3D Edits for Diffusion Models by Lifting Activations to 3D	2024
Karran Pandey , Paul Guerrero, Matheus Gadelha, Yannick Hold-Geoffroy, Karan Singh, Niloy Mitra CVPR Highlight .	
Juxtaform: interactive visual summarization for exploratory shape design	2023
Karran Pandey , Fanny Chevalier, Karan Singh ACM SIGGRAPH North America.	
Face Extrusion Quad meshes	2022
Karran Pandey , Jakob Andreas Baerentzen, Karan Singh ACM SIGGRAPH North America.	
A GPU Parallel Algorithm for Computing Morse-Smale Complexes	2022
Varshini Subhash, Karran Pandey , Vijay Natarajan IEEE Transactions on Visualization and Computer Graphics (TVCG).	
Morse Theory-based Segmentation and Fabric Quantification of Granular Materials	2022
Karran Pandey , Talha bin Masood, Saurabh Singh, Ingrid Hotz, Vijay Natarajan, Tejas Murthy Granular Matter 24, 27.	

GPU Parallel Computation of Morse-Smale Complexes

Varshini Subhash, **Karran Pandey**, Vijay Natarajan

IEEE Visualization Conference (VIS) (Short Paper).

An Integrated Geometric and Topological Approach for the Visual Analysis of Rossby Wave Packets

Karran Pandey, Joy Merwin Monteiro, Vijay Natarajan

Monthly Weather Review, 2020, 148 (8): 3139-3155.

CONFERENCE TALKS

ACM SIGGRAPH North America Emerging Technologies

Painting with 3D Gaussian Splat Brushes

August 2025
Vancouver, Canada

ACM SIGGRAPH North America Real-time Live!

Painting with 3D Gaussian Splat Brushes

August 2025
Vancouver, Canada

ACM SIGGRAPH North America

Painting with 3D Gaussian Splat Brushes

August 2025
Vancouver, Canada

ACM SIGGRAPH North America

Juxtaform: interactive visual summarization for exploratory shape design

August 2023
Los Angeles, USA

ACM SIGGRAPH North America

Face Extrusion Quad Meshes

August 2022
Vancouver, Canada

ACM SIGGRAPH North America (Labs Demo)

Face Extrusion Quad Meshes

August 2022
Vancouver, Canada

INDUSTRY EXPERIENCE

Adobe Research

Research Scientist Intern

Advisors: Paul Guerrero, Niloy Mitra, Matheus Gadelha, Yannick Hold-Geoffroy

- Image-to-Video Generation with Object-Focused Motion Diversity

A training-free pipeline for generating diverse object-focused motions with camera control using pretrained image-to-video diffusion models. Paper accepted to CVPR 2025.

June 2024 - September 2024

NVIDIA Toronto AI Lab

Research Scientist Intern

Advisor: Masha Shugrina

- Painting with 3D Gaussian Splat Brushes

A real-time 3D painting approach using brushes created from real-world gaussian splat captures. Paper accepted to ACM SIGGRAPH 2025 and live demo accepted to SIGGRAPH 2025 Real-time Live! and Emerging Technologies.

February 2024 - June 2024

Adobe Research

Research Scientist Intern

Advisors: Paul Guerrero, Niloy Mitra, Matheus Gadelha, Yannick Hold-Geoffroy

- 3D-aware edit handles for text-to-image diffusion models

A training-free pipeline for 3D-aware object editing using pretrained text-to-image diffusion models. Paper accepted to CVPR 2024 (Highlight).

June 2023 - September 2023

PROFESSIONAL ACTIVITIES

Reviewer

CVPR, SIGGRAPH, SIGGRAPH Asia, UIST, Eurographics

TEACHING

- Teaching Assistant at University of Toronto

Introduction to Computer Graphics (Lead), Design of Interactive Computational Media, Data Visualization

AWARDS

DGP Lansdale-Okino Graduate Fellowship	2023
5,000 CAD	

Department Of Computer Science Graduate Program Award	2022
5,971 CAD	

University of Toronto Fellowship, Faculty of Arts and Science	2022
1,381 CAD	